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Printer having sensor to detect remaining wound amount of label paper

Abstract

In accordance with an embodiment, a printer prints on labels attached to rolled paper at predetermined intervals. The printer includes a sensor that outputs a detection signal where a diameter of the rolled paper is a predetermined value or less. The printer detects, on the basis of the detection signal output from the sensor, a near-end state during an issuing operation of one label. Further, the printer determines, where the near-end state has been detected continuously for a predetermined number or more of labels, that a remaining amount of the rolled paper has been a predetermined amount.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2022-176302, filed on Nov. 2, 2022, the entire contents of which are incorporated herein by reference.
FIELD
(2) An embodiment to be described here generally relates to a printer.
BACKGROUND
(3) In the past, a printer that prints on rolled paper wound in a roll has been known. For example, a printer that prints on labels of rolled paper to which the labels are attached at predetermined intervals has been known. In this type of printer, the fact that the remaining amount of rolled paper is low is detected (hereinafter, referred to also as “remaining amount detection”) and a user is

notified of the fact.
(4) The remaining amount detection of rolled paper is performed using a sensor disposed to face a side surface of the rolled paper. Specifically, the sensor detects that the position of the outer peripheral surface of the rolled paper in the radial direction is a predetermined position. As a result, the printer detects that the winding amount of the rolled paper has decreased, i.e., the remaining amount of the rolled paper has been low.
(5) A printer that accurately performs remaining amount detection of rolled paper by holding down the rolled paper so as not to move in the radial direction during a printing operation of the printer has been proposed. However, this printer needs to have a structure for holding down the rotating rolled paper, and thus, the structure of the printer is complicated. For this reason, it is desired to be able to perform remaining amount detection of rolled paper with a simple structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic diagram showing a schematic configuration of a label printer according to an embodiment.
(2) FIG. **2** is a diagram showing part of rolled paper according to the embodiment.
(3) FIG. **3** is a diagram for describing movement of the rolled paper during an issuing operation of a label in the label printer according to the embodiment.
(4) FIG. **4** is a timing chart showing an example of an output of a near-end sensor in the label printer according to the embodiment.
(5) FIG. **5** is a block diagram showing a hardware configuration of the label printer according to the embodiment.
(6) FIG. **6** is a diagram showing a data configuration of a predetermined-number-of-times management table stored in a storage device of the label printer according to the embodiment.
(7) FIG. **7** is a diagram showing a data configuration of a number-of-steps management table stored in the storage device of the label printer according to the embodiment.
(8) FIG. **8** is a block diagram showing a functional configuration of a control unit of the label printer according to the embodiment.
(9) FIG. **9** is a flowchart showing remaining amount detection processing by the control unit of the label printer according to the embodiment.
(10) FIG. **10** is a flowchart showing remaining amount detection processing according to a modification of the control unit of the label printer according to the embodiment.
DETAILED DESCRIPTION
(11) In accordance with an embodiment, a printer includes: a holder; a printing device; a near-end sensor; and a controller. The holder rotatably supports rolled paper to which labels are attached at predetermined intervals. The printing device prints on the labels attached to the rolled paper fed out from the holder. The near-end sensor outputs a detection signal where a diameter of the rolled paper rotatably supported by the holder is a predetermined value or less. The controller is configured to detect, on the basis of the detection signal output from the near-end sensor, a near-end state during an issuing operation of one label. The controller is further configured to determine, where the near-end state has been detected continuously for a predetermined number or more of labels, that a remaining amount of the rolled paper has been a predetermined amount or less.
(12) Hereinafter, a printer according to an embodiment will be described in detail with reference to the drawings. In the drawings, the same reference symbols indicate the same or similar portions. Note that the embodiment is not limited by the following description. For example, although an example in which a label printer that prints on a label in an arbitrary operation mode is used as a printer will be described this embodiment, the embodiment is not limited thereto. The printer may

be a printer that prints on rolled paper to which no label is attached as long as a conveying motor that conveys rolled paper performs a reverse operation as in a modification described below.
(13) First, a schematic configuration of a label printer will be described. FIG. **1** is a schematic diagram showing a schematic configuration of a label printer according to an embodiment.
(14) In a label printer **1**, rolled paper **505** in which label paper **503** is wound in a roll is housed in a casing **2**. The rolled paper **505** is rotatably supported by a holder **30** (see FIG. **3**) described below. The rolled paper **505** includes a roll support **506** and the label paper **503** wound around the roll support **506**. The label printer **1** prints on the labels while drawing out the label paper **503** from the roll support **506**.
(15) The label paper **503** drawn out from the roll support **506** is sequentially conveyed in the direction indicated by arrows shown in FIG. **3** via a damper roller **50**. The damper roller **50** is attached to the casing **2** by a cantilever support structure in which one end thereof in the axial direction is attached to the casing **2**, similarly to a support shaft **31** of the holder **30** (see FIG. **3**). The damper roller **50** hangs the label paper **503** drawn out from the rolled paper **505** thereon to soften the impact applied to the rolled paper **505** at the moment when the label paper **503** lost its slack during the printing operation. Specifically, the damper roller **50** softens, when a conveying motor **40** is driven to convey the label paper **503** in the forward direction while the label paper **503** is loose, the impact applied to the rolled paper **505** at the moment when the label paper **503** lost its slack.
(16) As shown in FIG. **2**, the label paper **503** includes mounting paper **504** and a plurality of labels **502** attached to the mounting paper **504** at predetermined intervals. The plurality of labels **502** is attached to the mounting paper **504** at equal intervals with a predetermined label pitch "P". Note that the label pitch "P" is a distance between the tip of the label **502** in the conveying direction and the tip of the label **502** adjacent to the label **502** in the conveying direction. Note that the arrow in the figure indicates the conveying direction of the label paper **503**, which is a conveying direction during forward rotation of the conveying motor **40** described below (see FIG. **5**).
(17) As shown in FIG. **1**, the label printer **1** includes, inside the casing **2**, a conveying roller **11**, a platen roller **12**, a thermal head **13**, a label sensor **14**, a peeling guide **15**, a winding roller **16**, a peeling sensor **17**, and the damper roller **50**. Further, the label printer **1** includes, inside the casing **2**, a ribbon holding shaft **21**, a ribbon winding shaft **22**, and a guide shaft **23**.
(18) The conveying roller **11** includes a capstan roller **111** and two auxiliary rollers **112**. The label paper **503** drawn out from the roll support **506** is inserted between the capstan roller **111** and the auxiliary rollers **112**. The platen roller **12** is disposed at a position facing the thermal head **13**. The label paper **503** is inserted between the platen roller **12** and the thermal head **13**.
(19) The capstan roller **111** and the platen roller **12** are driven to rotate by the conveying motor **40** (see FIG. **5**). The conveying motor **40** is, for example, a forward and reverse rotatable stepping motor. The conveying motor **40** causes the capstan roller **111** and the platen roller **12** counterclockwise in the figure during forward rotation to convey the label paper **503** toward an outlet **3**. Further, the conveying motor **40** causes the capstan roller **111** and the platen roller **12** to rotate clockwise in the figure during reverse rotation to convey the label paper **503** in the direction opposite to the direction of the outlet **3**. In the following description, conveying the label paper **503** toward the outlet **3** will be referred to as "convey in the forward direction" and conveying the label paper **503** in the opposite direction will be referred to as "convey in the reverse direction" in some cases.
(20) The label printer **1** has a continuous issuing mode, a cut issuing mode, and a peeling issuing mode. The continuous issuing mode is an operation mode in which a set number of labels **502** are continuously printed and the label paper **503** is discharged from the outlet **3**. In the label paper **503** discharged from the outlet **3**, the plurality of printed labels **502** has been attached to the mounting paper **504**. In the continuous issuing mode, the conveying motor **40** rotates forward to convey the label paper **503** only in the forward direction.

(21) The cut issuing mode is an operation mode in which the printed labels **502** are cut into each piece. Specifically, the cut issuing mode is a mode in which the mounting paper **504** located between the labels **502** is cut by a cutter (not shown) while the label paper **503** to which the printed labels **502** are attached is discharged from the outlet **3**. As a result, the printed labels **502** are discharged one by one while being attached to the mounting paper **504**. After the cutter cuts the mounting paper **504**, the conveying motor **40** rotates in the reverse direction to convey the label paper **503** in the reverse direction and convey the label **502** to be printed next to the print start position.
(22) The peeling issuing mode is an operation mode in which the label **502** is peeled from the mounting paper **504**. Specifically, the peeling issuing mode is a mode in which most of the printed labels **502** are peeled from the mounting paper **504** and discharged from the outlet **3**. When some labels **502** attached to the mounting paper **504** are taken out by a user, the label printer **1** prints on the next label **502**. The conveying motor **40** rotates, when the peeled label **502** are taken out by the user, in the reverse direction to convey the label paper **503** in the reverse direction and convey the label **502** to be printed next to the print start position. Since the label paper **503** is conveyed in the reverse direction in the cut issuing mode and the peeling issuing mode, these modes are collectively referred to as a “reverse transfer mode” in some cases.
(23) The thermal head **13** is an example of a printing device that prints on the label **502** of the rolled paper **505** fed out from the holder **30**. That is, the thermal head **13** prints on the label **502** of the label paper **503** drawn out from the roll support **506**. The thermal head **13** is also capable of printing on the rolled paper **505** to which the label **502** is not attached. The thermal head **13** has a structure in which a plurality of heating elements is aligned. The thermal head **13** heats the heating elements corresponding to a print pattern to print on the label **502** of the rolled paper **505** sandwiched between the platen roller **12** and the thermal head **13**.
(24) Specifically, an ink ribbon **501** is inserted between the platen roller **12** and the thermal head **13**. The ink applied to the ink ribbon **501** is transferred to the label **502** of the rolled paper **505** by the heated thermal head **13**.
(25) The ink ribbon **501** is suspended between the ribbon holding shaft **21** and the ribbon winding shaft **22**. The ribbon holding shaft **21** winds the unused ink ribbon **501** in a roll. The ribbon winding shaft **22** is a shaft for winding the ink ribbon **501**. Further, the guide shaft **23** is a guide member for guiding the ink ribbon **501** suspended between the ribbon holding shaft **21** and the ribbon winding shaft **22** into a predetermined position. The ribbon winding shaft **22** is driven to rotate clockwise in the figure by a first drive motor (not shown) when printing on the label paper **503** and winds the ink ribbon **501**.
(26) Note that the thermal head **13** moves up and down by a moving mechanism (not shown) such as a solenoid. As a result, the label printer **1** is capable of switching between a state where the thermal head **13** is pressed against the platen roller **12** via the ink ribbon **501** and the rolled paper **505** and a non-pressure contact state where the thermal head **13** is away from the platen roller **12**. The thermal head **13** is pressed against the platen roller **12** via the ink ribbon **501** when printing on the rolled paper **505**. Further, the ribbon winding shaft **22** winds, during the printing, the ink ribbon **501** at a speed according to the conveying speed of the label paper **503** and stops the winding when the thermal head **13** enters the non-pressure contact state described above.
(27) The label sensor **14** is provided on a conveying path of the label paper **503** between the conveying roller **11** and the platen roller **12**. The label sensor **14** detects the tip portion of the label **502** in the conveying direction (hereinafter, referred to simply also as the “tip portion of the label **502**”) from the label paper **503**. As a result, the label sensor **14** is capable of detecting the label pitch “P” (see FIG. **2**). For example, the label sensor **14** can be realized by a transmissive sensor that includes a light-emitting element and a light-receiving device. The label sensor **14** detects the tip portion of the label **502** on the basis of the light reception level of the light-receiving device when conveying the label paper **503**.

(28) The label printer **1** calculates the position of the label **502** from the position of the tip portion of the label **502** detected by the label sensor **14** and conveys the label **502** to the print start position of the thermal head **13** in each operation mode.
(29) The peeling guide **15** peels the label **502** printed in the peeling issuing mode from the mounting paper **504**. The peeling guide **15** is formed in a V shape having two faces crossing each other at an acute angle. The peeling guide **15** bends the label paper **503** conveyed toward the outlet **3** to separate the mounting paper **504** and the label **502** from each other. While the mounting paper **504** from which the label **502** has been peeled is wound by the winding roller **16**, the label **502** peeled from the mounting paper **504** is discharged (issued) from the outlet **3** provided in the casing **2**.
(30) The winding roller **16** holds one end of the rolled paper **505** and winds the mounting paper **504** from which the label **502** has been peeled, in the peeling issuing mode. The winding roller **16** is driven to rotate by a second drive motor (not shown). For example, the second drive motor causes, when printing on the label paper **503**, the winding roller **16** to rotate counterclockwise in the figure to wind the mounting paper **504** from which the label has been peeled. The winding roller **16** does not hold the rolled paper **505** and is not used in the continuous issuing mode and the cut issuing mode.
(31) The peeling sensor **17** is installed in the vicinity of the outlet **3** and detects the presence or absence of the labels **502** most of which have been peeled from the mounting paper **504** in the peeling issuing mode. The peeling sensor **17** can be realized by, for example, a transmissive sensor that includes a light-emitting element and a light-receiving device.
(32) When the peeling sensor **17** detects the label **502**, the label printer **1** temporarily stops the conveyance and printing of the label paper **503**. When a user takes up the label **502** from the outlet **3**, the peeling sensor **17** detects that the label **502** is not present. The label printer **1** restarts the conveyance and printing of the label paper **503** in the case where the peeling sensor **17** has detected that no label is present.
(33) Specifically, the label printer **1** conveys, in the case of restating the printing, the label paper **503** in the reverse direction by a predetermined amount in order to return the next label following the peeled label to the print start position of the thermal head **13**. The label printer **1** prints, when the conveyance in the reverse direction is completed, on the next label and issues the printed label from the outlet **3**.
(34) Next, the remaining amount detection of the rolled paper **505** will be described. FIG. **3** is a diagram for describing movement of the rolled paper **505** during an issuing operation of a label. In FIG. **3**, the holder **30** that rotatably supports the rolled paper **505** includes the support shaft **31**. The support shaft **31** is attached to the casing **2** by a cantilever support structure in which one end thereof in the axial direction is attached to the casing **2**. The support shaft **31** is inserted through the roll support **506** of the rolled paper **505** to rotatably support the rolled paper **505**. Note that although not shown, a guide that restrains the movement of the rolled paper **505** in the axial direction is provided in the other end of the support shaft **31**.
(35) The label printer **1** includes a near-end sensor **41**. The near-end sensor **41** is provided at a position facing a side surface of the rolled paper **505** and detects a near-end state where the outer peripheral surface of the rolled paper **505** in the radial direction has been at a predetermined position. In other words, the near-end sensor **41** detects a near-end state where the diameter of the rolled paper **505** has been a predetermined amount or less by drawing out the label paper **503**. The near-end sensor **41** can be realized by, for example, a reflective sensor that includes a light-emitting element and a light-receiving device.
(36) In the support structure of the rolled paper **505** described above, the label paper **503** has a trajectory deviated from a virtual line “LPL” when being drawn out from the roll support **506** and passing through the damper roller **50**. Specifically, the label paper **503** has a curl and has a trajectory indicated by a broken line in the figure due to the difference in strength of the curl

between the portion to which the label **502** is attached and the portion to which no label is attached. In more detail, both end portions of the label **502** are recessed inward.

(37) For this reason, at the point indicated by "A" in the figure, the rolled paper **505** supported by the support shaft **31** is pressed by the label paper **503** drawn out from the roll support **506** (portion located at the end portion of the label **502**). The event that the rolled paper **505** is pressed by the label paper **503** drawn out from the roll support **506** occurs for each issuing operation of one label. The rolled paper **505** is pressed by the label paper **503** drawn out from the roll support **506** to move from a normal portion (hereinafter, referred to as "roll") in some cases. In this case, the near-end sensor **41** detects the near-end state even if the winding amount of the rolled paper **505** is not a predetermined amount or less.

(38) The pressing force on the rolled paper **505** by the label paper **503** drawn out from the roll support **506** is not constant due to the degree of the curl of the label **502** and the like. For this reason, the rolled paper **505** does not roll in some cases while the remaining amount is the predetermined amount or more and the mass thereof is large, even if it is pressed by the label paper **503** drawn out.

(39) FIG. **4** is a timing chart showing an example of an output of the near-end sensor **41** in the continuous issuing mode. In the example shown in FIG. **4**, in the case where the remaining amount of the rolled paper **505** is the predetermined amount or more, the pressing force applied to the rolled paper **505** during the issuing operation of a first label is small and the rolled paper **505** does not roll. For this reason, the near-end sensor **41** does not output the near-end state and does not output a detection signal (high level signal).

(40) Note that in FIG. **4**, an issuing operation of one label **502** (issuing operation of one label) is, for example, an operation of the label printer **1** from the time point when the label **502** reached the print start position to the time point when the next label **502** is conveyed to the print start position. In other words, the issuing operation of one label **502** is the operation of the thermal head **13**, the conveying motor **40**, and the like while the label paper **503** is conveyed by the amount of the label pitch "P" in the continuous issuing mode. In the following description, the period in the issuing operation of one label **502** will be referred to also as "one label issuing period".

(41) In the example shown in FIG. **4**, in the case where the remaining amount of the rolled paper **505** is the predetermined amount or more, the pressing force applied to the rolled paper **505** in the one label issuing period for the second label is large and the rolled paper **505** rolls. For this reason, the near-end sensor **41** detects the near-end state when the rolled paper **505** is pressed by the label paper **503** to roll and outputs a detection signal. Note that in the one label issuing period for the second label, after the rolled paper **505** is pressed by the label paper **503** at the point "A" (see FIG. **3**), the rolled paper **505** returns to the normal portion, and thus, the near-end sensor **41** has returned to a state where it does not output a detection signal.

(42) In this example, the near-end sensor **41** detects the near-end state only in the one label issuing period for the second label while N labels **502** are continuously issued. In other words, the near-end sensor **41** detects the near-end state in the one label issuing period for the second label although the remaining amount of the rolled paper **505** is the predetermined amount or more.

(43) In the case where the remaining amount of the rolled paper **505** is the predetermined amount or less, the near-end sensor **41** detects the near-end state when the rolled paper **505** is at the normal portion and continues to output a detection signal. Even if the rolled paper **505** is pressed by the label paper **503** to roll in the continuous issuing mode, the near-end sensor **41** detects the near-end state and thus outputs a detection signal. In this regard, the label printer **1** according to this embodiment determines, in the case where the near-end sensor **41** has detected the near-end state continuously for a predetermined number or more of labels **502** in the continuous issuing mode, that the remaining amount of rolled paper has been low. Note that also in other operation modes, similarly, it is possible to prevent erroneous detection of the remaining amount detection of the rolled paper **505** due to that the rolled paper **505** is pressed by the label paper **503**.

(44) FIG. **5** is a block diagram showing a main hardware configuration of the label printer **1**. The label printer **1** includes a controller **100**, a storage device **200**, the thermal head **13**, the label sensor **14**, the peeling sensor **17**, the conveying motor **40**, the near-end sensor **41**, a cutter motor **42**, a display device **43**, an operation device **44**, and a communication device **45**. The controller **100**, the storage device **200**, the thermal head **13**, the label sensor **14**, the peeling sensor **17**, the conveying motor **40**, the near-end sensor **41**, the cutter motor **42**, the display device **43**, the operation device **44**, and the communication device **45** are connected to each other via a bus **46** or the like.
(45) The controller **100** includes a central processing unit (CPU) **101**, a read only memory (ROM) **102**, and a random access memory (RAM)**103**. The CPU **101**, the ROM **102**, and the RAM **103** are connected to each other via the bus **46**.
(46) The CPU **101** controls the operation of the entire label printer **1**. The CPU **101** is an example of a processor. The ROM **102** stores various programs such as the program to be used for driving the CPU **101**, and various types of data. The RAM **103** is used as a work area of the CPU **101**. The CPU **101** develops the various programs and the various types of data stored in the ROM **102** and the storage device **200**. The CPU **101** operates in accordance with the control program that is stored in the ROM **102** and the storage device **200** and developed into the RAM **103**, thereby executing various types of control processing of the label printer **1**.
(47) Further, the RAM **103** includes a print data section **1031**. The print data section **1031** stores print data, the instructed number of copies to be printed, and the like received from an external apparatus such as a personal computer (hereinafter, referred to as a PC) **60** (see FIG. **8**). Examples of the print data include character data and image data to be printed on the label **502**. The instructed number of copies to be printed is the number of the labels **502** on which the print data is to be printed.
(48) The storage device **200** includes a storage medium such as a hard disk drive (HDD) and a flash memory and retains the stored content even if the power is cut off. The storage device **200** stores a control program **201**, a label pitch section **202**, a predetermined-number-of-times management table **203**, and a number-of-steps management table **204**.
(49) The control program **201** is a program for realizing a function of acquiring print data from an external apparatus such as the PC **60**, a function of driving the thermal head **13** and the conveying motor **40** to print the print data on a medium (the label **502**, the rolled paper **505**, or the like), a function of detecting the label pitch “P” on the basis of the output of the label sensor **14**, a function of performing the remaining amount detection on the basis of the output of the near-end sensor **41**, and the like. The control program **201** includes various other control programs for operating the label printer **1**.
(50) The label pitch section **202** shown in FIG. **5** stores the label pitch of the rolled paper **505** to be used. The label pitch section **202** stores the label pitch detected by the label sensor **14**. For example, the controller **100** of the label printer **1** conveys, when the rolled paper **505** is newly supported by the holder **30**, the label paper **503** by a predetermined amount, detects the label pitch of the rolled paper **505** by the label sensor **14**, and stores the detected label pitch in the label pitch section **202**. The label pitch section **202** may store the label pitch input by operating the operation device **44** by a user.
(51) The predetermined-number-of-times management table **203** is a table for storing the number of consecutive times (predetermined number of times) of the near-end state to be used for determining the remaining amount detection in accordance with the label pitch. The predetermined-number-of-times management table **203** can be arbitrarily set by a user. FIG. **6** is a diagram showing a data configuration of the predetermined-number-of-times management table **203**. The predetermined-number-of-times management table **203** stores a label pitch range and a predetermined number of times in association with each other.
(52) The label pitch range is a range of the label pitch of the rolled paper **505** to be used. In this embodiment, as the label pitch range, three ranges of 50 mm or less, larger than 50 mm and 100

mm or less, larger than 100 mm are set.

(53) The predetermined number of times is the number of consecutive times of the near-end state to be used for determining the remaining amount detection. In other words, the label printer **1** determines, in the case where the near-end state has been detected a predetermined number of times continuously in one label issuing period, that the remaining amount detection has been performed, i.e., the remaining amount of the rolled paper **505** has been detected as a predetermined amount or less. By changing the predetermined number of times depending on the range of the label pitch, it is possible to prevent the determination of remaining amount detection from taking more time than necessary in the case where the rolled paper **505** having the long label pitch "P" is used.

(54) The number-of-steps management table **204** is a table in which the number of steps of the conveying motor **40** necessary in one label issuing period is set for each label pitch. FIG. **7** is a table showing a data configuration of the number-of-steps management table **204**. The number-of-steps management table **204** stores the label pitch and the number of steps for one label in association with each other.

(55) The label pitch "P" is the label pitch of the rolled paper **505** to be used. The label pitch to be registered in the number-of-steps management table **204** may be registered in advance when the label printer **1** is shipped, or data stored in the label pitch section **202** may be used. Alternatively, the label pitch to be registered in the number-of-steps management table **204** may be input from the operation device **44** when a user uses the rolled paper **505**.

(56) The number of steps for one label is the number of steps for driving the conveying motor **40** in one label issuing period. The number of steps for one label to be registered in the number-of-steps management table **204** may be registered in advance when the label printer **1** is shipped or may be calculated by the controller **100** when the label pitch is registered. In this case, since the conveying distance of the conveying motor **40** for one step is determined, the controller **100** is capable of calculating the number of steps for one label corresponding to the label pitch.

(57) With reference to FIG. **5** again, the hardware configuration of the label printer **1** will be described. The thermal head **13**, the label sensor **14**, the peeling sensor **17**, the conveying motor **40**, and the near-end sensor **41** are as described above.

(58) The cutter motor **42** drives a cutter (not shown) that cuts the label paper **503** discharged from the outlet **3**. Specifically, the cutter motor **42** drives, in the cut issuing mode, the cutter while the label paper **503** to which the printed labels **502** have been attached is discharged from the outlet **3**. As a result, the printed labels **502** are discharged one by one while being attached to the mounting paper **504**.

(59) The display device **43** includes, for example, a liquid crystal panel and is provided on the outer surface of the casing **2**. The display device **43** displays various types of information. For example, the display device **43** displays, in each operation mode, the instructed number of copies to be printed and the number of prints of the label **502** stored in the print data section **1031**. Further, the display device **43** displays, when the remaining amount detection is performed, information indicating that the remaining amount of the rolled paper **505** is a predetermined amount or less, in each operation mode. The display device **43** is an example of a notification device that notifies that the remaining amount of the rolled paper **505** is a predetermined amount or less. Note that the notification device may include a speaker for audio output.

(60) The operation device **44** includes, for example, a touch panel provided on the surface of the display device **43**. The operation device **44** is operated by a user of the label printer **1** to input various types of information to the controller **100**. For example, the operation device **44** inputs information for instructing the start of printing, information for setting an operation mode, or the like to the controller **100**.

(61) The communication device **45** is an interface for communicating with an external apparatus such as the PC **60**. The controller **100** is connected to the external apparatus via the communication device **45** and thus is capable of transmitting/receiving information (data) to/from the external

apparatus.
(62) Next, a functional configuration of the label printer **1** will be described. FIG. **8** is a block diagram showing a main functional configuration of the label printer **1**. When the CPU **101** operates in accordance with the control program stored in the ROM **102** and the storage device **200**, the controller **100** functions as an acquisition unit **1001**, an input unit **1002**, a mode setting unit **1003**, a printing control unit **1004**, a label pitch detection unit **1005**, a near-end state detection unit **1006**, a remaining amount determination unit **1007**, and a display control unit **1008**. Note that these functions may be realized by hardware such as a dedicated circuit.
(63) The acquisition unit **1001** acquires print data. Specifically, the acquisition unit **1001** receives, from the PC **60**, print data such as character data and image data to be printed on the label **502**. Further, the acquisition unit **1001** receives, from the PC **60**, the instructed number of copies to be printed, which is associated with the print data. The acquisition unit **1001** stores the acquired print data and the acquired instructed number of copies to be printed in the print data section **1031**.
(64) Various types of information are input to the input unit **1002** from the label sensor **14**, the peeling sensor **17**, the near-end sensor **41**, and the operation device **44**. For example, in each operation mode, position information of the label **502** to be conveyed is input to the input unit **1002** from the label sensor **14**. The position information of the label **502** includes information indicating the positions of the front end portion and the rear end portion of the label **502** in the conveying direction. The controller **100** performs printing control in each operation mode on the basis of the position information of the label **502** input to the input unit **1002**. Further, the controller **100** detects, when new rolled paper **505** is loaded into the holder **30**, for example, the label pitch of the rolled paper **505** on the basis of the position information of the label **502** input to the input unit **1002**.
(65) In the peeling issuing mode, peeling information indicating whether or not the printed label **502** is located at the outlet **3** is input to the input unit **1002** from the peeling sensor **17**. The controller **100** controls, on the basis of the peeling information input to the input unit **1002**, the thermal head **13** and the conveying motor **40** to perform printing control in the peeling issuing mode.
(66) In each operation mode, detection information indicating whether or not the rolled paper **505** has been detected is input to the input unit **1002** from the near-end sensor **41**. Further, various types of information are input to the input unit **1002** from the operation device **44**. For example, setting information for setting an operation mode is input to the input unit **1002** from the operation device **44**.
(67) The mode setting unit **1003** sets the operation mode of the label printer **1**. Specifically, the mode setting unit **1003** sets, on the basis of the setting information input to the input unit **1002**, an operation mode such as the continuous issuing mode, the cut issuing mode, and the peeling issuing mode as the operation mode of the label printer **1**.
(68) The printing control unit **1004** controls the thermal head **13**, the conveying motor **40**, the cutter motor **42**, and the like in accordance with the operation mode set by the mode setting unit **1003** to print on the rolled paper **505**.
(69) Specifically, the printing control unit **1004** controls the thermal head **13**, the conveying motor **40**, and the like in the continuous issuing mode to continuously print the label **502** and discharge the label paper **503** from the outlet **3**. Further, the printing control unit **1004** controls the thermal head **13**, the conveying motor **40**, the cutter motor **42**, and the like in the cut issuing mode to cut the printed labels **502** into each piece and issue it. Further, the printing control unit **1004** controls the thermal head **13**, the conveying motor **40**, and the like in the peeling issuing mode to peel the printed label **502** from the mounting paper **504** and issue it.
(70) The label pitch detection unit **1005** detects the label pitch “P” of the rolled paper **505**. Specifically, the label pitch detection unit **1005** detects, on the basis of the position information of the label **502** input to the input unit **1002**, an interval between one end in the conveying direction of

the label **502** attached to the mounting paper **504** and one end in the conveying direction of the next label **502**. The label pitch detection unit **1005** stores the detected label pitch in the label pitch section **202**.

(71) The near-end state detection unit **1006** detects the near-end state on the basis of the output of the near-end sensor **41**. Specifically, the near-end state detection unit **1006** detects, when detection information indicating that the near-end sensor **41** does not detect the rolled paper **505**, i.e., a high-level detection signal, is input to the input unit **1002**, that the rolled paper **505** is in the near-end state.

(72) The remaining amount determination unit **1007** determines, in the case where the near-end state detection unit **1006** detects the near-end state continuously for a predetermined number or more of labels **502** in an arbitrary operation mode, that the remaining amount of the rolled paper **505** is a predetermined amount or less. In the case where the near-end state has been detected a predetermined number of times continuously in one label issuing period in each operation mode, the remaining amount determination unit **1007** detects that the remaining amount of the rolled paper **505** is the predetermined amount or less. For example, in the case where the label pitch "P" of the rolled paper **505** is 50 mm or less, the remaining amount determination unit **1007** determines, when the output of the near-end sensor **41** shown in FIG. **4** is a high level continuously from the issuing of a first label to the issuing of a sixth label, that the remaining amount of the rolled paper **505** is the predetermined amount or less.

(73) The display control unit **1008** displays various types of information on the display device **43**. For example, the display control unit **1008** displays, in the case where the remaining amount determination unit **1007** has determined that the remaining amount of the rolled paper **505** is the predetermined amount or less, information indicating this fact on the display device **43**.

(74) Next, remaining amount detection processing executed by the controller **100** of the label printer **1** will be described. FIG. **9** is a flowchart showing the remaining amount detection processing by the controller **100** of the label printer **1**. Note that the label pitch "P" of the rolled paper **505** has been stored in the label pitch section **202** before the remaining amount detection processing is executed. Further, the remaining amount detection processing is executed during an issuing operation in an arbitrary operation mode.

(75) In Step S**1**, the controller **100** reads the label pitch of the rolled paper **505** from the label pitch section **202**. Subsequently, in Step S**2**, the controller **100** refers to the predetermined-number-of-times management table **203** (see FIG. **6**) to set the predetermined number of times corresponding to the read label pitch as a cumulative counter value N. For example, in the case where the label pitch read from the label pitch section **202** is 50 mm or less, the cumulative counter value N is set to six. Similarly, in the case where the read label pitch is larger than 50 mm and 100 mm or less, the cumulative counter value N is set to four. In the case where the read label pitch is larger than 100 mm, the cumulative counter value N is set to two.

(76) Subsequently, in Step S**3**, the controller **100** serves as the printing control unit **1004** to recognize the first step of the drive signal of the conveying motor **40** in one label issuing period. In Step S**4**, the controller **100** serves as the remaining amount determination unit **1007** and the near-end state detection unit **1006** to determine, on the basis of the output of the near-end sensor **41**, whether or not the near-end state has been detected. In other words, the controller **100** serves as the remaining amount determination unit **1007** to determine whether or not detection information indicating that the rolled paper **505** is not detected has been input to the input unit **1002** from the near-end sensor **41**.

(77) In the case where the controller **100** (near-end state detection unit **1006**) has detected the near-end state (Yes in Step S**4**), the processing of the controller **100** proceeds to Step S**5**. In Step S**5**, the controller **100** sets the detection counter to "1". In the case where the controller **100** (near-end state detection unit **1006**) has not detected the near-end state (No in Step S**4**), the processing of the controller **100** proceeds to Step S**6**. In Step S**6**, the controller **100** sets the detection counter to "0".

For example, the controller **100** stores the detection counter value set in the RAM **103**.
(78) Subsequently, in Step S**7**, the controller **100** determines whether or not the drive signal in the final step of one label issuing period has been output. In the case where the drive signal in the final step of one label issuing period has not been output (No in Step S**7**), the processing of the controller **100** returns to Step S**4**. As a result, the near-end state detection unit **1006** of the controller **100** is capable of detecting that the near-end state has been reached at arbitrary timing in one label issuing period.
(79) In the case where the drive signal in the final step of one label issuing period has been output (Yes in Step S**7**), the processing of the controller **100** proceeds to Step S**8**. In Step S**8**, the controller **100** determines whether or not the detection counter value is "1". In other words, the controller **100** determines whether or not the near-end state has been detected in one label issuing period. In the case where the detection counter value is "1" (Yes in Step S**8**), the processing of the controller **100** proceeds to Step S**9**. In Step S**9**, the controller **100** adds "1" to the value of the cumulative counter. The value of the cumulative counter is stored in, for example, the RAM **103**.
(80) Subsequently, in Step S**10**, the controller **100** serves as the remaining amount determination unit **1007** to determine whether or not the value of the cumulative counter has reached the set value "N". In the case where the value of the cumulative counter has reached "N" (Yes in Step S**10**), the controller **100** serves as the remaining amount determination unit **1007** to determine that the remaining amount of the rolled paper **505** has been a predetermined amount or less. The processing of the controller **100** proceeds to Step S**11**. In Step S**11**, the controller **100** serves as the display control unit **1008** displays, on the display device **43**, information indicating that the remaining amount of the rolled paper **505** has been small, which is a predetermined amount or less. The display device **43** displays, for example, prompting replacement of the rolled paper **505** with new one.
(81) Subsequently, in Step S**12**, the controller **100** determines whether or not the labels **502** corresponding to the instructed number of copies to be printed included in the print data acquired by serving as the acquisition unit **1001** have been issued. In other words, the controller **100** determines whether or not an issuing operation for one label has been executed for the instructed number of copies to be printed. When the labels **502** corresponding to the instructed number of copies to be printed have been issued (Yes in Step S**12**), the controller **100** finishes the remaining amount detection processing. In the case where the labels **502** corresponding to the instructed number of copies to be printed have not been issued (No in Step S**12**), the processing of the controller **100** returns to Step S**3**. The controller **100** continues the remaining amount detection processing until the labels **502** corresponding to the instructed number of copies to be printed are issued.
(82) Note that in the case where the value of the detection counter is not "1" in Step S**8** (No in Step S**8**), the processing of the controller **100** proceeds to Step S**13**. In Step S**13**, the controller **100** sets the value of the cumulative counter to "0". The processing of the controller **100** proceeds to Step S**12**. Further, in the case where the value of the cumulative counter is not "N" in the processing of Step S**10** (No in Step S**10**), the processing of the controller **100** skips Step S**11** and proceeds to Step S**12**.
(83) By the remaining amount detection processing described above, the label printer **1** is capable of preventing erroneous detection of the remaining amount detection of the rolled paper **505** due to that the rolled paper **505** is pressed by the label paper **503**, without providing a structure for holding down the rolled paper **505**, or the like.
(84) Next, a modification of the remaining amount detection processing will be described. The modification prevents erroneous detection of the remaining amount detection of the rolled paper **505** due to the tension applied to the label paper **503** during an issuing operation of a label. For example, when conveying the label paper **503** or when cutting the label paper **503** by a cutter, tension is applied to the label paper **503**, which causes the rolled paper **505** to roll in some cases. In

the modification, erroneous detection of the remaining amount detection is prevented by performing the remaining amount detection while tension is not applied to the label paper **503** as much as possible.

(85) FIG. **10** is a flowchart showing a modification of the remaining amount detection processing by the controller **100** of the label printer **1**. The remaining amount detection processing according to the modification is executed in the reverse transfer mode such as the cut issuing mode and the peeling issuing mode.

(86) In Step S**21** shown in FIG. **10**, the controller **100** determines whether or not the conveying motor **40** rotates forward. The controller **100** performs the determination processing of Step S**21** on the basis of whether or not the conveying motor **40** is caused to rotate forward by serving as the printing control unit **1004**. In the case where the conveying motor **40** does not rotate forward (No in Step S**21**), the processing of the controller **100** returns to Step S**21**. In the case where the conveying motor **40** rotates forward (Yes in Step S**21**), the processing of the controller **100** proceeds to Step S**22**. In Step S**22**, the controller **100** determines whether or not the conveying motor **40** has reversed. In other words, the controller **100** determines whether or not the conveying motor **40** has rotated in the reverse direction. The controller **100** performs the determination processing of Step S**22** on the basis of whether or not the printing control unit **1004** has caused the conveying motor **40** to rotate in the reverse direction.

(87) In the case where the conveying motor **40** has not reversed (No in Step S**22**), the processing of the controller **100** returns to Step S**22**. In the case where the conveying motor **40** has reversed (Yes in Step S**22**), the processing of the controller **100** proceeds to Step S**23**. In Step S**23**, the controller **100** serves as the remaining amount determination unit **1007** and the near-end state detection unit **1006** to determine whether or not the near-end state has been detected. In other words, the controller **100** serves as the remaining amount determination unit **1007** and the input unit **1002** to determine whether or not detection information indicating that the rolled paper **505** is not detected has been input from the near-end sensor **41**. The controller **100** serves as the remaining amount determination unit **1007** to determine whether or not it is the near-end state while the conveying motor **40** rotates in the reverse direction and the label paper **503** is loose, i.e., the label paper **503** does not pull the rolled paper **505** supported by the holder **30**.

(88) When the controller **100** serves as the near-end state detection unit **1006** to detect the near-end state (Yes in Step S**23**), the processing of the controller **100** proceeds to Step S**24**. In Step S**24**, the controller **100** serves as the display control unit **1008** to display, on the display device **43**, information indicating that the remaining amount of the rolled paper **505** has been small, which is a predetermined amount or less.

(89) Subsequently, in Step S**25**, the controller **100** determines whether or not the labels **502** corresponding to the instructed number of copies to be printed included in the print data acquired by serving as the acquisition unit **1001** have been issued. When the labels **502** corresponding to the instructed number of copies to be printed have been issued (Yes in Step S**25**), the controller **100** finishes the remaining amount detection processing. In the case where the labels **502** corresponding to the instructed number of copies to be printed have not been issued (No in Step S**25**), the processing of the controller **100** returns to Step S**21**. The controller **100** continues the remaining amount detection processing until the labels **502** corresponding to the instructed number of copies to be printed are issued.

(90) Note that in the case where the controller **100** does not serve as the near-end state detection unit **1006** to detect the near-end state in the processing of Step S**23** (No in Step S**23**), the processing of the controller **100** skips Step S**24** and proceeds to Step S**25**.

(91) By the remaining amount detection processing described above, the label printer **1** is capable of preventing erroneous detection of the remaining amount detection of the rolled paper **505** due to the tension applied to the label paper **503**, without providing a structure for holding down the rolled paper **505**, or the like.

(92) As described above, the label printer **1** according to this embodiment includes the holder **30**, the thermal head **13**, the near-end sensor **41**, and the controller **100**. The holder **30** rotatably supports the rolled paper **505** that includes the mounting paper **504** to which the labels **502** are attached at predetermined intervals being wound. The thermal head **13** prints on the label **502** of the rolled paper **505** fed out from the holder **30**. The near-end sensor **41** outputs a detection signal when the outer peripheral surface of the rolled paper **505** in the radial direction is at a predetermined position, the rolled paper **505** being rotatably supported by the holder **30**. The controller **100** serves as the near-end state detection unit **1006** to detect, on the basis of the output of the near-end sensor **41**, the near-end state during an issuing operation of one label. Further, the controller **100** serves as the remaining amount determination unit **1007** to determine, in the case where the near-end state detection unit **1006** has detected the near-end state continuously for a predetermined number or more of labels, that the remaining amount of the rolled paper **505** has been a predetermined amount or less.
(93) As a result, the label printer **1** is capable of preventing erroneous detection of the remaining amount detection of the rolled paper **505** due to that the rolled paper **505** is pressed by the label paper **503**, without particularly providing a structure for holding down the rolled paper **505**.
(94) Further, in the label printer **1** according to this embodiment, the controller **100** serves as the near-end state detection unit **1006** to determine, on the basis of the drive signal of the conveying motor **40** that conveys the rolled paper, that an issuing operation of one label is being performed.
(95) As a result, the label printer **1** is capable of determining that an issuing operation of one label is being performed, with a simple configuration. Therefore, it is possible to simplify the configuration for preventing erroneous detection of the remaining amount detection of the rolled paper **505**.
(96) Further, in the label printer **1** according to this embodiment, the controller **100** serves as the label pitch detection unit **1005** to detect an interval between one end in the conveying direction of the label **502** attached to the mounting paper **504** and one end in the conveying direction of the next label **502**. The controller **100** serves as the remaining amount determination unit **1007** to set the predetermined number of times described above in accordance with the detected interval.
(97) As a result, the label printer **1** is capable of preventing the determination of remaining amount detection from taking more time than necessary in the case where the rolled paper **505** having the long label pitch “P” is used.
(98) In addition, the label printer **1** according to this embodiment includes the holder **30**, the thermal head **13**, the near-end sensor **41**, the conveying motor **40**, and the controller **100**. The holder **30** rotatably supports the rolled paper **505** wound in a roll. The thermal head **13** prints on the rolled paper **505** fed out from the holder **30**. The near-end sensor **41** outputs a detection signal when the outer peripheral surface of the rolled paper **505** in the radial direction is at a predetermined position, the rolled paper **505** being rotatably supported by the holder **30**. The conveying motor **40** rotates forward to convey the rolled paper **505** in the conveying direction and rotates in the reverse direction to convey the rolled paper **505** in the direction opposite to the conveying direction. The controller **100** serves as the near-end state detection unit **1006** to output a detection signal when the outer peripheral surface of the rolled paper **505** in the radial direction is at a predetermined position, the rolled paper **505** being rotatably supported by the holder **30**. Further, the controller **100** serves as the remaining amount determination unit **1007** to determine, in the case where the near-end state detection unit **1006** has detected the near-end state while the conveying motor **40** does not rotate forward, that the remaining amount of the rolled paper **505** has been a predetermined amount or less.
(99) As a result, the label printer **1** is capable of preventing erroneous detection of the remaining amount detection of the rolled paper **505** due to the tension applied to the label paper **503** in the reverse transfer mode, without particularly providing a structure for holding down the rolled paper **505**, or the like.

(100) Further, in the label printer **1** according to this embodiment, the controller **100** displays, on the basis of the determination result obtained by serving as the remaining amount determination unit **1007**, information for informing that the remaining amount of the rolled paper **505** has been a predetermined amount or less on the display device **43**.
(101) As a result, the label printer **1** is capable of informing a user of that the remaining amount of the rolled paper **505** has been a predetermined amount or less.
(102) Note that in the embodiment described above, the control program executed by the label printer **1** may be recorded on a computer-readable recording medium such as a CD-ROM and provided. Further, the control program executed by the label printer **1** according to the embodiment described above may be provided by storing the control program on a computer connected to a network such as the Internet and downloading it via the network. Further, the control program may be provided via a network such as the Internet.
(103) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A printer, comprising: a holder that rotatably supports rolled paper in which label paper with a plurality of labels attached thereto at predetermined intervals is wound; a printing device that prints on the labels attached to an unwound part of the label paper fed out from the holder; a near-end sensor that outputs a detection signal where an outer peripheral surface of the rolled paper rotatably supported by the holder in the radial direction has reached a predetermined position; and a controller configured to detect, on a basis of the detection signal output from the near-end sensor, a near-end state of the rolled paper during an issuing operation of each of a set number of labels printed by the printing device, and determine, where the near-end state has been detected continuously for a predetermined number or more of labels out of the set number of labels, that a remaining wound amount of the label paper of the rolled paper is a predetermined amount or less.
2. The printer according to claim 1, wherein the near-end sensor is provided at a position facing a side surface of the rolled paper.
3. The printer according to claim 1, further comprising a conveying motor that conveys the label paper of the rolled paper, the controller being further configured to determine, on a basis of a drive signal of the conveying motor, that an issuing operation of one label is being performed.
4. The printer according to claim 1, further comprising a label sensor that detects a tip portion of a label in a conveying direction.
5. The printer according to claim 4, wherein the label sensor is provided on a conveying path through which the unwound part of the label paper of the rolled paper fed out from the holder is conveyed.
6. The printer according to claim 4, wherein the controller is further configured to detect, on a basis of detection information of the label sensor, an interval between an end of the label in the conveying direction and an end of a next label in the conveying direction.
7. The printer according to claim 4, wherein the controller is further configured to set the predetermined number in accordance with the detected interval, and determine, where the near-end state has been detected continuously for the predetermined number or more of labels, that the remaining wound amount of the rolled paper has been the predetermined amount or less.
8. A printer, comprising: a holder that rotatably supports rolled paper in which label paper with a

plurality of labels attached thereto at predetermined intervals is wound; a printing device that prints on the labels attached to a part of the label paper fed out from the holder; a near-end sensor that outputs a detection signal where an outer peripheral surface of the rolled paper rotatably supported by the holder in the radial direction has reached a predetermined position; a conveying motor that, during an issuing operation, rotates forward to feed out the part of the label paper from the holder and convey the part of the label paper in a conveying direction and rotates in a reverse direction to convey the part of the label paper in a direction opposite to the conveying direction for printing of subsequent labels, a tension not being applied to the part of the label paper while the conveying motor does not rotate in the forward direction; and a controller configured to detect, on a basis of the detection signal output from the near-end sensor, a near-end state of the rolled paper during the issuing operation, and determine, where the near-end state has been detected while the conveying motor does not rotate forward, that the remaining wound amount of the label paper of the rolled paper is a predetermined amount or less.
9. The printer according to claim 8, further comprising a notification device, the controller being further configured to notify, on a basis of a result of the determination, that the remaining wound amount of the rolled paper is the predetermined amount or less via the notification device.